Socialist Planning in the Era of Climate Change

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Introduction

The case for socialism is often laid out on moral grounds. It is said that socialism will put people before profits; that it will end the unjust servitude of labor to the upper classes; and that it will bring about equal opportunities for individuals to flourish.[1] This cannot be achieved by a mere redistribution of wealth, since modern societies rest upon the mobilization of labor with large-scale interdependent productive assets – such as mines, railways, power plants and hospitals. These assets are not amenable to direct redistribution, and therefore the promises of socialism can be realized only through their *collective* utilization. We will therefore use a broad definition of a *socialist economy* as one in which the working population exerts political control over the utilization of these assets.[2]

Yet, the desirability of socialism is not sufficient to render it feasible. A socialist economy cannot simply be wished into existence. Its *economic* feasibility rests on the possibility of building it out of elements from the existing capitalist environment within the lifespan of a given population. At the same time, the *political* feasibility of socialism requires it to address problems impeding the material well-being for a vast majority. We cannot expect to build mass movements with members committed to undertaking the risks of economic transformation, or defending its outcome, unless socialism can demonstrate successive gains over capitalism.[3] We take these economic and political requirements to be the elementary *feasibility conditions* of socialism. They relegate all designs predicated on unconstrained abundance or widespread altruistic behaviors to the realm of science fiction and utopian speculation. Such designs lack strategic relevance for socialist movements aiming to build institutions that operate in a world constrained by material incentives, divisions of labor, and territorially bounded nation states.

In the advanced countries of the 1930s, a feasible socialism had to offer full employment and modernizing investments amidst capitalist stagnation and mass unemployment. In the 1960s, by contrast, it had to offer gains beyond the concessions won by organized labor under state-led, full employment growth paths. In the 2030s, the tightening of resource constraints through climate changes, geopolitical conflict and demographic decline places new requirements on what socialism must plausibly deliver. Current projections estimate that the effects of climate changes alone will depress average real incomes across most of the world over the coming twenty-five years due to declining labor and agricultural productivity as well as rising costs of environmental damages.[4] Worse still, the livelihoods and essential needs of the masses are put at stake as the severity of heat waves, forest fires and floods continually increases. The problems of adapting to and mitigating these conditions alter the coordinates of socialist politics.

The scale is certainly nontrivial. Adaptation requires massively fortifying critical supply systems; the construction of defenses against rising sea-levels; and the transfer of coastal cities towards

[1] See for instance Cohen (2009); Roemer (2017); Vrousalis (2021).

[2] On one hand, this definition is consonant with a materialist conception in which economic systems are distinguished by their means of extracting and appropriating labor. On the other hand, it avoids utopian idealist conceptions of socialism by underscoring that the socialist character of an economy is a matter of degree, since the forms of political control can be more or less direct and encompassing.

[3] For discussions about the economic and political feasibility of socialism, see Nove (1991) and Przeworski (1985), respectively.

[4] See e.g. Kotz et al. (2024).

the inland. Mitigation requires a transformation towards an energy system based on stable non-fossil sources; an electrification of the transportation system; a modernization of the housing stock; the rewilding of forests; and so on. Stemming the demographic decline is likely to require a substantial transfer of investments towards supporting human reproduction.

Relative to a deteriorating capitalist environment, a feasible socialist economy must possess the capacity to *efficiently* secure a range of resources required for the well-being of its population, while *limiting* the quantities of extracted critical resources and emitted waste products. We will refer to this as the *Viable Allocation Problem*. It disqualifies the focus of the 'degrowth' paradigm on downscaling material throughput, since it risks imposing austerity on the masses.[5]

In this paper, we begin by arguing that socialist economies, in which production largely relies on decentralized market exchange, cannot solve the allocation problem due to informational and incentive reasons. We conclude that in these circumstances, the feasibility conditions for socialism necessitate *planning*, which one of its early Scandinavian theorists argued is motivated by a need to coordinate immediate and future economic actions.[6] For instance, the winding down of fossil fuel production in ten years must be coordinated with the ability to secure electrification and living standards beyond that period to be feasible.[7] Thus planning is a particular *type* of economic regulation to meet politically defined *goals* over time. If our assessment above is correct, it means that planning must once again become a central, albeit not exhaustive, element of socialist economic policy.

Historically, the promise of planning carried most weight during periods of depression, destruction and development. An illustration is provided by the post-war program adopted by the Swedish labor movement in 1944: "We do not have to resign ourselves to unemployment, and we can abolish destitution, provided that the productive power, which the war revealed us to possess, is used with similar expedience for peacetime purposes. The main task before us is to *coordinate economic activities* into a planned economic system, so that the labor force and the material assets are utilized for efficient production all the time. Such coordination ought to take place under the direction of society and should be so arranged that private interests are subordinated to the goals for which we strive *collectively*."[8]

Planning, however, has its own set of fundamental information and incentive problems, which we highlight in the case of the Soviet-type economies. With these problems in mind, we outline a minimal feasible structure for democratic economic planning to tackle the Viable Allocation Problem. Building on ideas from economists who were deeply engaged with the problems of Soviet-type planning, we then describe a set of instruments and institutional mechanisms that can overcome the information and incentive problems.

Limits of Market Allocation

Imagine a socialist economy in which capitalist ownership of productive assets has been replaced by collective ownership, but where resources are priced in a decentralized manner by worker-managed firms and distributed according to their acquired purchasing power.[9] How will the retained market allocation of critical resources – energy, fuels, food crops and so on – tackle the Viable Allocation Problem? The Austrian Marxist Otto Neurath argued already at the beginning

[5] See Hickel and Sullivan (2024) for an alternative focus on needs.

[6] See Svennilson (1965), who was pursuing research on planning in the West in the 1930s.

[7] Thus disrupting the fossil energy base – either by legal means or by sabotage as advocated by Malm (2021) – *without* the planned construction of an alternative one is a politically infeasible strategy.

[8] Arbetarrörelsens efterkrigsprogram – De 27 punkterna med motivering, https://www.marxistarkiv.se/sverige/socialdemokratin/efterkrigsprogrammet.pdf. Emphasis added.

[9] Such proposals include, with varying emphasis, Roemer (1994); Schweickart (2002); Beggs (2022) and, more recently, Beggs et al. (2026).

of the previous century that such allocation squanders society's ultimate resources: both human labor and the gifts of nature. He illustrated this by the example of coal, a finite resource that was put to waste for use in heating rather than in the production of useful chemicals where it could not readily be substituted.[10] Decentralized price adjustments, and the resulting changes to incomes, have no way of anticipating and adjusting to the collective consequences of depleted critical resources or accumulated waste.

Around the same time as Neurath drew his conclusion, neoclassical economists, such as Arthur Pigou, realized that producers' failure to take into account the 'external' effects of their decisions could lead to resource allocations that would leave all worse off than if they were to coordinate their decisions. The solution was to regulate market allocation by either adjusting the pricing or limiting physical quantities, such as freshwater consumption, carbon emissions, and land usage.

Let us consider the use of pricing instruments first. How should a regulatory body adjust prices of, say, fossil fuels to direct their overall usage to meet collective needs within specified physical constraints? A fuel tax or subsidy would affect everything from investments in railways to the provision of medical supplies. The regulatory body would need to gather *information* from millions of interdependent firms to even begin anticipating these effects. Suppose, by some decree, that firms were required to disclose such detailed information; then there would be *no incentive* for them to report it accurately. Whatever decisions firms make, their bottomline is determined by net incomes from sales. Since their reports to the regulatory body would affect their incomes, they would not disclose any information that would be deleterious to this interest. Thus the regulatory body would be acting upon partial and even distorted information. A change in the form of ownership, from capitalist to collective, does not, by itself, relocate the relevant knowledge: it still resides with whoever manages a firm's day-to-day production, whose own interest in its continued operation – assessed, under market exchange, by its net income from sales – remains distinct from the collective interest the regulatory body is meant to serve.[11] Thus price regulations lead either to persistent overshoot of resource constraints or falling short of meeting collective needs. An overshoot is only registered and adjusted by a regulator after the resource has been extracted or the waste emitted. For a cumulative limit such as a carbon budget, or for investments with long gestation, that correction comes too late: the limit is exactly what cannot be exceeded and then repaired. But the same conclusions hold if we instead considered using quantitative caps on, say, fossil fuels.[12]

The problem persists also in more sophisticated proposals that socialize investment while retaining competitive product markets. In the model of Beggs et al. (2026), building on Schweickart (2002), public banks finance firms on the basis of cash-flow forecasts and risk-adjusted terms, while the central bank caps aggregate investment. Each bank can fund projects without observing the sum of every other bank's claims on the same finite resources, and a cap on the total value of investment is not a cap on any particular resource: nothing in this arrangement supplies any bank with the information needed to see whether the local investment decisions add up to a jointly feasible use of aggregate resources. Worse, misjudged investments are identified only through disappointing returns and reviewed at refinancing, that is, *after* the resources they drew on have already been spent. Thus resources can easily be squandered and stipulated limits on resource extraction or waste emissions exceeded. Avoiding this requires that investment decisions be coordinated within aggregate physical limits, not corrected individually after they fail – a task to which we return below.

[10] Neurath (1919), cited in Uebel (2005).

[11] As we point out below, the Soviet-type economies encountered a very similar alignment-of-interests problem despite public ownership of enterprises.

[12] Even within idealized economic models, the uncertainties that a regulatory body faces render the very choice of effective instruments ambiguous, see Weitzman (1974).

We therefore conclude that market allocation is faced with information and incentive problems that undermine the ability to solve the Viable Allocation Problem.[13]

Recently, Benanav proposed that transactions across collectively owned enterprises should be made public through an electronic ledger or database.[14] This still requires a complementary mechanism to appropriately steer their joint production: Consider an enterprise producing intermediate products for other enterprises. Suppose it has been informed that switching to a more expensive supplier would reduce the carbon footprint or that reducing its current production would free up constrained resources to meet an increased demand for insulin. Faced with such information it would nevertheless determine its production, purchases and prices in accordance with its own interests to receive revenues for its operations.

It is precisely this lack of coordination that motivates *planning* as a means of solving the Viable Allocation Problem. Since the 1990s, there has been a limited but gradually renewed interest in planning.[15] While its proponents have widened the imagination of socialist thought, we believe they have not adequately tackled the fundamental information and incentive problems that arise specifically in the coordination of large economic organizations. We therefore turn to these problems as they emerge in the Soviet-type economies.[16]

Soviet-Type Planning – Prospects and Problems

Socialists were first forced to confront the challenges of planning in the destructive context of wartime mobilization, especially where they seized power in Russia and parts of Germany. The First World War not only altered the coordinates of socialist politics, it profoundly shaped socialist economic theory. Specifically, the wartime economy of imperial Germany provided a model for Lenin.[17] After devastating wars, Soviet planning institutions emerged from the needs of industrial development, starting with the electrification of a large agrarian country.

While the Soviet political structure limited democratic deliberation and contestation over plan goals, its form of planning was far from an unmitigated disaster, contrary to current liberal ideology. In the 1920s, Soviet economists were opening new frontiers of research by making rapid advances in e.g. methods of national accounts and the theories of economic growth and investment – until they were stifled by the political culture under Stalin. While no major capitalist market economy could escape the devastation of the Great Depression, the Soviet economy exhibited an extraordinary expansion of output and investment during this period. From 1928 to 1940, it achieved a scale of industrial transformation that had taken several decades to achieve in other countries. As a consequence, its people experienced a dramatic transformation of their livelihoods and living standards.[18] While first improvised out of necessity, planning had become a decisive factor by the Second World War. The Soviet economy could mobilize its production more rapidly than its mortal enemy, despite starting off from a much weaker industrial base.[19] Even capitalist states of the United States and United Kingdom suspended market coordination

[13] Problems are compounded by firms with large employment or share of output that risk losses under market regulation. Their economic weight would exert pressure to receive exemptions or cut favourable deals.

[14] See Benanav (2025a,b), which takes inspiration from Devine (1988) and his subsequent work.

[15] Representative examples include Devine (1988), Cockshott and Cottrell (1993), Albert and Hahnel (1991) as well as the special issues of *Science & Society* on socialist visions, published in 1992, 2002, 2012 and 2022. For a comparison, see Legault and Tremblay-Pepin (2021). More recent proposals discussing socialist planning include Phillips and Rozworski (2019), Vettese and Pendergrass (2022), Cockshott et al. (2022), Hannah (2024) and Benanav (2025a,b). See also Gindin (2018), Grünberg (2025) and Ackerman (2026) for a critical discussion.

[16] We thank Moshé Machover, Adaner Usmani, Sam Gindin, David Laibman and Petter Nilsson for their constructive feedback on this paper.

[17] Smolinski (1969).

[18] See Gregory and Sailors (2003) and Allen (2003).

[19] Harrison (2000).

and installed comprehensive planning institutions for the war effort. Well into the 1950s, it was feared among Western elites that the USSR would overtake the US. Its exceptional economic modernization process certainly bolstered the appeal of using planning techniques in the post-war reconstruction of France, the Netherlands, and Japan as well as parts of post-independence Africa, Asia, and South America.

Soviet-type planning involved three institutional layers: central planning authorities, industrial/regional ministries, and individual enterprises. Long-term development goals were politically determined by the center and operationalized as annual *production targets* for a wide range of products. These were arrived at through an iterative process of *material supply balancing*:[20] Each year, the center would announce aggregate targets, after which the enterprises responded with their estimated quantities of inputs required to meet these initial targets. The center and ministries would then attempt to balance the internal supply and demand figures until settling on the official production targets. The fulfillment of these targets was one of several economic criteria used to *evaluate* enterprises. While capable of rapid industrialization, the planning system was also riddled with persistent problems of rigidities, inefficiencies, product quality and shortages, which were only partially compensated by horizontal coordination within gray and black markets.[21] Some problems were of course historically specific, such as those arising from the enormous diversion of resources towards military defense through both hot and cold wars. But fundamentally there were mutually reinforcing problems of *information* and *incentives* that arise in large economic organizations.[22]

Firstly, the *downstream* information was inadequate. While the balancing method was a rather ingenious innovation for coordinating production under severe informational limits, it could not arrive at consistent plans in which the internal resource requirements and supplies would balance to meet the intended targets. The annual procedure was not responsive enough to efficiently adapt to rapidly changing conditions. Moreover, it reinforced structural rigidities by setting growth-driven targets not only for final products but also those used up in production. Importantly, its costing methods provided no reliable means of evaluating how different economic alternatives put internal resources to use in achieving plan goals. This led to a squandering of resources.

Secondly, the *upstream* information was both inaccurate and incomplete. While the planning process affected tens of millions of distinct product-types, ranging from electricity to footwear, they had to be selected and crudely aggregated into a few thousand prioritized types to be processed at the center layer. This masked information about efficient or wasteful practices at the lower layers. The procedure also had few means of eliciting timely and local information about production options, technical problems, quality grades or sudden opportunities which could improve the feasibility or efficiency of the plans. The pooling of such information at the center was necessarily partial.[23]

Thirdly, the *alignment* of interests between the policy makers at the center layer and the decision makers at lower layers failed. On the one hand, the central plans determined both the output expected from the enterprises and the resources allocated to them. On the other hand, the center depended on information provided by the *very same* enterprises to arrive at the plans in the first place.[24] The incentives therefore drove enterprises and ministries to understate production

[20] Montias (1959) and Manove (1971) provide a characterization of the method of material balances.

[21] There is a vast literature on the problems of Soviet-type economies. For an overview, see for instance Dobb (1970), Nove (1986, 1991, 1992), Ellman (1973, 1978, 2014), Kornai (1980), Dyker (1985), Gregory (1990), Eatwell et al. (1990), Ericson (1991) and Allen (2003).

[22] Information and incentive problems became an active research program following the so-called socialist calculation debates. See, for instance, Hurwicz (1973); Johansen (1977, 1978) and Radner (1987).

[23] This was a key argument in Hayek (1945).

[24] In the economics literature, this is referred to as a 'principal-agent problem', see e.g. Gregory (1990), Harrison (2005).

possibilities and report exaggerated output data so as to moderate and meet central targets. At the same time, enterprises facing uncertainties in the supply allocation found it rational to report inflated input requirements and hoard resources in order to secure their production. Wasteful enterprises lacked an effective budget constraint, since the center was reluctant to wind them down. Overall demand for inputs and investment goods could then easily outpace the growth of supplies, thereby cementing persistent shortages. The process of formulating consistent plans was also undermined by opaque negotiations with sectoral groups that bargained for their local interests.[25]

A series of reform efforts aimed at addressing several of the above challenges led to mixed results. Improved computerized planning methods were pioneered in Hungary by the late 1950s and found later success in a few areas, such as production scheduling and investment allocations. However, there were both technological limits to process the requisite economic data and institutional limits to elicit reliable information in the first place. By the end of the 1960s, the leading Soviet mathematical economist Viktor Novozhilov concluded that the institutional information and incentive problems held back growth by more than any mathematically superior planning method could ever have recovered.[26]

Institutional reforms were pioneered in Yugoslavia, which had achieved a high degree of political independence from the USSR due to the role of its domestic forces in defeating fascist occupation. In 1950, it rejected Soviet-type planning in favor of a system of self-managed enterprises. However, it unleashed some of the highest unemployment levels in Europe and cemented deep regional inequalities. More dramatically, the market-based reforms introduced under Gorbachev wrecked the planning mechanisms of the USSR and sent it into terminal decline. Both cases ended with disastrous social consequences. On the other hand, Hungary, where substantial elements of decentralized mechanisms were introduced, fared far better. Together with the developed economies of Czechoslovakia and East Germany, they continued to improve material living standards, but could never close the gap to their European neighbors. The persistent gaps undermined the political feasibility of these economies. It is indicative that when East Germans were polled four years after unification about their new economic conditions, they responded overwhelmingly that commodity and leisure supplies were superior under welfare-state capitalism.[27]

None of the reform efforts succeeded in resolving or offsetting the fundamental problems of information and incentives in Soviet-type planning. Even if it could use later advances in information technologies to tackle some of the information problems, they could not overcome the institutional problems. The ruptural paths through which communist parties seized and monopolized power for their survival led to resistance against institutional innovations that implied relinquishing control. Furthermore, lacking feasible technological alternatives, the established planning process required a bureaucratic apparatus to operate, which concentrated economic power in the planning agencies who were also unwilling to give it up. In the USSR, the system had after all delivered a dramatic economic transformation and defeated an industrial war machine, which only cemented conservatism against fundamental institutional changes.

Socialist Planning – a Feasible Minimum

[25] See for instance Kőhegyi and Kovács (2022).

[26] See citation in Ellman (1973, p. 40) and remark in Novozhilov (1970, footnote 7). Together with Kantorovich and Nemchinov, Novozhilov was instrumental in reviving mathematical economics in the USSR after the death of Stalin.

[27] Poll by Institut für Demoskopie Allensbach (Spring 1994) cited in Therborn (1995). The poll also revealed that the same East Germans overwhelmingly thought that safety, child rearing, the supply of interesting jobs and the sense of community, were superior in their former economy despite its deficiencies.

In the 1950s, Oskar Lange, who was chairing a council charged with reforming Soviet-type planning in post-war Poland, understood that tackling the problems of information and incentives in planning required a different starting point than an administrative war economy.[28] Rejecting the Soviet model leads to a series of questions:

1. What should be the scope of planning?
2. What are its structural constraints and openings?
3. How should its collective goals be determined democratically?

We will consider each question in turn, aiming to identify a *minimal* set of institutional changes that can enable a polity to tackle the Viable Allocation Problem. Specifically, we advocate a public utilities approach to socialist planning that *fuses* political and productive power for economic transformation.

Scope of planning

The reproduction of *any* industrialized economy, socialist or otherwise, involves a planned sector, a market sector, households and an external sector. The households supply labor power, while the external sector supplies products that are impossible or too costly to produce domestically. The planned sector utilizes assets that are collectively owned and we will refer to its units of production simply as *enterprises*. This sector commonly includes, among other things, education, healthcare and many utilities. For concreteness, we will assume public ownership, but other legal forms are conceivable.

Since the requirements for running a hospital or constructing a railway line are very different, the management structure of each enterprise must vary with the nature of work and its internal division of labor. Even across similar enterprises, however, workers' preferences may differ in how their workplaces should be managed. Some may prefer highly participatory forms, others may settle on simply selecting their managers. It is a matter to be stipulated in the statutes of each enterprise. Our only assumption here is that workers of an enterprise have an interest in keeping it *operational*.

The appropriate division between the planned and market sectors has been a longstanding debate in socialist economics. The classical socialist perspective was to take aim at the 'commanding heights' of the economy but we think it fails to provide adequate guidance in modern-day economies. Not all needs of society are to be collectively defined. There is a very large set of *individual* needs and preferences for consumer products that play no meaningful role in democratic deliberation.[29] On the other hand, society must collectively deliberate over *common* physical constraints, such as limited quantities of fresh water consumption, area of land use, or barrels of crude oil produced, which do impinge on these needs. To address the asymmetry between individual needs and common constraints, we adopt a *public utilities* perspective, which is concerned with the provision of public goods and the 'external' effects of economic decisions.

Utilities are meant to supply goods and services that are considered 'essential' or 'vital' in some economic sense. In an industrialized economy it becomes clear that certain product types – say, electrical power, steel, cement and so on – enter directly or indirectly into the production of every product and are therefore *essential* for the overall economic process.[30] This is fundamentally a functional classification dependent on the overall structure of production, independently of

[28] Lange (1958).

[29] Schweickart (2006) skewers the idea that individuals should submit periodic private consumption plans to a central planning process.

[30] The special role of such basic products in capitalism was first studied formally by the Italian economist Piero Sraffa (Sraffa 1960), who was building on prior work by Quesney and Marx. See also Cockshott and Zachariah (2006) and Zachariah and Cockshott (2020).

how a product is valued by any political process. Without power generation, transportation, or water, there is no chance of securing a viable economy at large.

Similarly, there are *vital* products which are directly and indirectly required to meet the collective needs of the working population. This too is a functional classification, but of a different kind: vital products are identified by the needs they meet rather than by their place in the technical structure of production. Crucially, *which* needs count as *collective* – as opposed to personal needs left to individual choice – is a political determination, open to democratic contestation and revision over time. Collective needs in most modern societies will, for instance, encompass healthcare, housing, commuting services and food crops[31]; which would therefore be treated as vital and hence subject to public provision.

From a public utilities perspective then, the planned sector should include workplaces that generate the *bulk of society's essential and vital products*. In the US economy, for instance, the people who keep the lights on, grow the food, run the hospitals, mine the ore, pour the concrete, move the freight, amount to something like half the workforce.[32] That is the scale of what a public utilities approach proposes to bring under collective direction: not the whole economy, and not a sliver of it, but the machinery everything else depends on.

The functional asymmetry between planned and market sectors achieves two things. On the one hand, the planned sector directs the supply chain of essential products which *erodes* market dependence and *imposes* physical constraints on production in the market sector. For instance, it can limit the supply of fossil fuels to prioritized sectors, while offering investment goods that accelerate electrification. Consumer demand in the market sector is also constrained by the products under public provision. For instance, if transportation is made an accessible public good, then the urban demand for private cars will decline. For the planned sector to become dominant, it should become self-sustaining in terms of its domestic inputs. This is not market socialism simply run at half scale. The market sector is not a separate domain of independent resource allocation – its essential inputs, investment credit and physical limits are all set by the planned sector – but a bounded arena for personal consumption goods and services.

On the other hand, the planning process is *relieved* of the task of exploring and satisfying individual consumer preferences. Such diverse preferences can change rapidly in response to supply changes and are rarely possible to consistently specify ex-ante in any detail. Instead, the task is laid upon firms in the market sector, such as restaurants, clothing manufacturers and car repair shops, who *purchase* their essential inputs from the planned sector.[33] A monetary circuit between firms and the planned sector must thus be facilitated.[34] In so far as public banks provide the main line of credit in the economy, capitalist ownership in the market sector is eroded by extending credits to cooperatively or municipally owned firms. In other words, the

[31] Buffer stocks in food crops – public accumulation of surpluses and releases during shortages or price spikes – have served to stabilize staple food prices in the market sector; see Weber et al. (2026).

[32] More specifically, it would include a significant portion of workers in the energy sector, transportation, logistics, telecommunications, manufacturing, construction, mining, agriculture as well as health care and basic education.

[33] Grossman (1963) pointed out challenges arising in the separation of market and planned sector. Specifically, the greater the dependency of the latter on the former, the more the unpredictability of market-driven production threatens the internal consistency of the plan. Our proposed functional asymmetry resolves this separability problem directly, since products serving individual consumption can be left to market coordination without threatening planning consistency.

[34] Prices for inputs sold to the market sector can be pegged to their shadow price, so that a firm's cost reflects the input's opportunity cost, leaving little room to arbitrage. When this risks price instability for market-facing products, quantitative buffers can absorb transient scarcity while keeping the sale price stable, since a buffer release keeps the quantity available matched to demand at that price rather than forcing rationing (see Weber et al. (2026)). A sustained rather than transient shift in scarcity then surfaces as a deliberate repricing decision by the sectoral boards, or as increased priority for producing or importing the buffered good, rather than as a gap left open for arbitrage.

asymmetry between planned and market sectors enables the former to regulate the development of the latter by *leveraging its political and productive power* towards collectively defined goals.

Structural constraints and openings for planning

The external sector imposes fundamental constraints on this power. First, no industrialized economy can dispense with imports without a major reduction of living standards. Second, socialist planning institutions must adapt to an external sector that is overwhelmingly capitalist. To presume otherwise only begs the question, as it assumes what is yet to be established: large-scale socialist economies. This means that the planned sector is constrained by the need to maintain a balance of payments with the external sector, ultimately financing imports through exported products. However, the scale of the planned sector can be leveraged when procuring imports from international markets. The geopolitical conjuncture in the capitalist world economy therefore sets varying constraints for domestic elements of socialist planning across countries.

Organizing for structural changes within a national capitalist economy risks triggering the decline of confidence among capitalist investors. This would threaten both employment and real income growth, as well as aggravate problems of deficits and balance of payments. The uncertain costs and duration of such transitional turbulence would quickly disintegrate mass support for building socialist institutions, even if their material welfare were to benefit in the long run. In other words, as long as the material welfare of the working population is tightly coupled to concessions made under capitalist growth, the obstacles to structural changes remain formidable.[35] These conditions of political feasibility must guide any transitional strategy, rather than invocations of magical ruptures.

Our wager, however, is that the conditions are being altered by climate changes, geopolitical conflict and demographic decline: Their effects increasingly impinge on economic development in a manner that *necessitates political interventions to secure economic viability.* On the one hand, the increased frequency and scope of political interventions has the effect of re-opening areas of contestation over economic structures. On the other hand, it blunts the threat of declining capitalist investment in the market sector – since interventions to secure economic reproduction are not conditioned on business confidence[36] – and potentially splits capitalist interests as different fractions of capital are unevenly exposed to the unfolding problems. While it is unlikely to bring about a breakdown and contestation over state power similar to cases in the twentieth century, it does open a political space in which a planned sector can expand control over specific, strategically pivotal nodes of production and infrastructure. Any feasible transitional path would require wielding the asymmetric productive power of the planned sector to erode its market dependence, while simultaneously delivering improvements in material welfare, including securing access to employment, energy, water, food supplies, housing, and transportation.[37] This approach could be characterized as a sectoral erosion of capitalist power.

Coordinating the joint production of essential and vital products necessitates the expansion of collective institutional capacities, as did the wartime mobilizations of the twentieth century. On the above wager, we have two reasons to believe that the structural changes can be directed by politically organized coalitions of working people within the essential and vital sectors. On the one hand, workers in the essential sectors are endowed with a disruptive potential which gives them disproportionate leverage over political interventions in structural crises. Obstructions in the essential production processes would paralyze the large parts of the economy within a short period of time. On the other hand, they would collectively possess the productive power to erode market dependence and direct economic development towards collectively defined needs. Indeed,

[35] A classic analysis of these structural obstacles is given in Przeworski (1985).

[36] This is most apparent in national emergencies, pandemics and wars.

[37] One strategic reason for maintaining a market sector for individual consumption products is to reduce the risks of suddenly disrupting consumption on a mass scale.

the sectors that historically conferred maximum political leverage – manufacturing, mining, construction and transport – form a core of what we identified above as essentials.[38]

All of this depends, of course, on developing the appropriate form of *political organization*, which remains the central open strategic issue for socialists in this century.

Collective goals of planning

How much crude oil should the country import next year? How many housing units should the planned sector build? Such collective goals should be deliberated and contested democratically among the people who will live with the answers. We suggest two institutions for this: a *viability assembly* that sets quantitative boundaries, and an *elected process* for determining the direction of economic development, with plans carried out through sectoral boards. First, the assembly is tasked to determine and revise broad *quantitative goals* that aim to secure a sustainable reproduction of the population over time. This includes required quantities of certain prioritized supplies, such as drinking water, electrical power and grains, but also limits on quantities of specific waste products, land usage or utilization of fresh water supplies.[39] In these cases, quantitative goals are superior to monetary instruments.[40]

The participation of groups of scientific advisors in the process would be vital, but the assembly must consist of stakeholders who face the consequences of their own decisions as citizens, workers, and consumers – unlike the heads of existing economic ministries and agencies. Only then can the costs and benefits of decisions (and indecisions) be shared fairly. There are several conceivable mechanisms for selecting representatives among stakeholders, including sortition which minimizes the distortion of the assembly's class character.[41] Regardless of selection mechanism, a central viability assembly determines certain aggregate goals, which will need to take into account information and demands from regional assemblies rooted in local communities.

Second, the plan's *direction*, which the assembly's boundaries do not fix, must also be set. To do so, Lange argued, planning institutions must determine the division of resources directed towards investments and consumption, respectively, over time.[42] Specifically, aggregate *sectoral budget allocations* towards, say, the health care and power supply sectors could subsequently be materialized as hospital wards and power plants.[43]

We envisage that this allocation is determined by the election of *economic programs* proposed by coalitions of political parties, which organize different interests and articulate competing visions of development. Some programs may prioritize allocations for high-speed railway transportation, while others may seek expanded housing with communal luxuries or the rewilding of some regions. It is also through this process that different strategies for adaptation to and mitigation of climate changes are politically debated. Elected programs periodically specify budget allocations designated for *investment* and *provision*, respectively, across sectors. This determines the division of the planned sector's final product.

The budget allocations towards investments and provision are managed by corresponding *sectoral boards*, in analogy to ministries and their regional subdivisions. Their key role is to grant and oversee *budgeted projects* that involve orders from one or more enterprises. For instance, the transportation investment board may grant a railway construction project in which steel is ordered from a steel plant. The power investment board may grant investments to modernize a

[38] See for instance Usmani (2018).

[39] It will also include setting viable limits on the balance of payments in trade over a given time horizon.

[40] See Weitzman (1974).

[41] For an alternative to electoral selection, see Burnheim (1985), Cockshott and Cottrell (1993) and Van Reybrouck (2016).

[42] Lange (1958).

[43] Around 10-20 sectors is a manageable division used in national accounts today.

power plant. *Investment projects* are proposed by a multiplicity of sources, be it municipalities, enterprises or professional organizations. *Provision projects* are set up between external parties and enterprises. For instance, a health council may order insulin from a producer under a health provision project or a bakery in the market sector may contract electrical power supplies under a power provision project.

The structure of democratic decision-making over collective plan goals is illustrated in Fig. 1. While it provides a means of directing resources to meet collective needs, it is underspecified for solving the Viable Allocation Problem: The institutions lack a mechanism ensuring that final orders and enterprises' own production remain consistent with the boundaries set by the assembly, and an incentive structure for enterprises to meet collective needs efficiently.

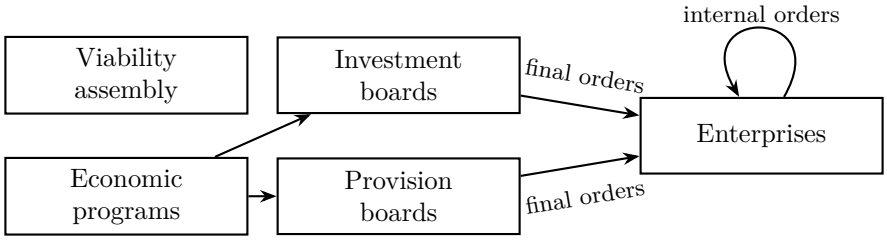


Figure 1: Collective goals and production in the planned sector. The viability assembly determines aggregate quantitative boundaries, while elected economic programs determine the allocation of funds to sectoral investment and provision boards. The boards grant project plans through which final orders are placed with enterprises. The enterprises place internal orders among themselves to obtain inputs for production.

Anticipatory Regulation – Instruments and Mechanisms

In 1957, the economic council that Lange headed recommended granting enterprises greater autonomy. While the reforms in Poland were quickly rolled back, similar reforms were implemented in 1968 in Hungary, abolishing central directive control rather immediately across a very large part of the economy. Yet this coordinative mechanism was not replaced by market price mechanisms to any greater extent. The economist Janos Kornai observed that the real economy nevertheless continued to function without any shock and formulated a research program dedicated to understanding how.[44]

Under the imperative to stay operational, an enterprise will aim to meet its contractually committed orders, striving to maintain its inventories by means of corrective changes to production and internal orders. This engenders an *autonomous* regulation process by interdependent enterprises, which has two desirable properties for the purpose of planning. First, it is capable of sustaining their joint production in the short run as well as overcoming unforeseen shocks that may arise from a wide range of reasons: from broken machines, delivery errors and incorrect orders, to extreme climate events such as floods, fires and droughts.[45] Second, it requires neither price adjustments nor central instructions, only self-managed adjustments in response to incoming orders and changes in inventory levels. Given their information about local economic opportunities, enterprises are best suited to explore their own mix of production inputs and outputs they wish to offer to potential users. Thus in the face of uncertainties, interdependent production systems can be regulated autonomously in the short run. This insight is the starting point for our proposed planning process.

[44]He did so by eschewing conventional economic analysis, using instead systems and control theory pioneered by Wiener (1948) and Simon (1952). See Kornai and Martos (1973), Kornai and Martos (1981) and Martos (1990). This theoretical perspective has also recently been rediscovered by Steblev and Kallis (2025).

[45]To use some examples cited in Ackerman (2026).

However, autonomous regulation cannot solve the Viable Allocation Problem. First, it is merely locally *reactive* to changes in the overall collective goals. If projects aim to rapidly change the energy supply mix, the process has no means of jointly securing essential products in a timely manner. This would require that each enterprise can internally represent how future collective goals propagate though the interdependent production system to affect its own production.[46] Lacking such capacities, the reactive nature induces delays or cycles with shortages. Second, autonomous regulation has no internal mechanism to prioritize the *allocation* of finite resources and productive capacities. If fossil fuel imports are limited by geopolitical conflict or environmental concerns, the process cannot bring about an allocation of such scarce quantities to satisfy collective goals efficiently. Indeed, since the process is driven by each enterprise's own imperative to remain operational, each unit will act to protect its own production and orders, even where this conflicts with a collectively prioritized use of the same scarce inputs elsewhere in the economy.

To tackle these limitations of autonomous regulation brings us back to problems of information and incentives, which we now set out to address. In this light, planning is to be understood as a complementary process of *anticipatory regulation* that coordinates immediate and future economic action. Its function is to facilitate the formulation of feasible collective goals by taking into account their anticipated requirements and steering the behaviour of enterprises towards an efficient use of resources to meet these goals.

Rolling plan horizons

In the Soviet-type economies, periodic plans separated plan formation and execution across time, which meant that when, say, a harvest comes in short, the shortfall ripples through the economy for months before a plan adapts to it. Already in the late 1930s, economists proposed *rolling plans* to fix exactly this mismatch between how fast a plan can change and how fast the world does.[47] That is, plans are formulated as a sequence of overlapping horizons as illustrated in Figure 2. The goals in each plan horizon are *incremental* in relation to those settled in the past overlapping plan horizon. Thus if in the previous plan horizon the power generation for a future month was contracted to be 100 GWh, then the new plan horizon only allows for a certain amount of additional power generation for the same month.

This makes it possible to adapt anticipated goals to changing circumstances as new information becomes available, while leaving the autonomous regulation by enterprises intact to react to local opportunities and unforeseen changes. By layering continual autonomous regulation with rolling anticipatory regulation, resource coordination can be achieved without succumbing to large delays or increasingly disruptive system-wide plan revisions.[48]

The key upstream information used for anticipatory regulation is elicited in the very process of autonomous regulation, where the enterprises' orders and inventories are managed electronically as in present-day procurement and inventory systems. Specifically, their orders, employment of labor and inventory changes are stored in a *public database* managed by a public coordination agency. We will now show how this structured information can be used to arrive at jointly feasible plan goals and an appropriate costing of resources for each new plan horizon.

[46] This is a consequence of the internal model principle discovered by control theorists. A local feedback control in a production networks must essentially generate unknown dynamics of all impinging exogenous signals, see Isidori (2025).

[47] Some pioneering work was in Svennilson (1938), later recounted in English in Svennilson (1965); see also Lundberg (1972).

[48] In the language of systems and control theory, fast enterprise-level feedback control is complemented by a lower-rate feedforward control implemented by a receding horizon method that is fed future final orders and quantitative boundaries, see Matni et al. (2024). The incremental revision of goals and the absorption of local shocks by autonomous regulation avoids the instability, or "schedule nervousness," that Ackerman (2026) raises.

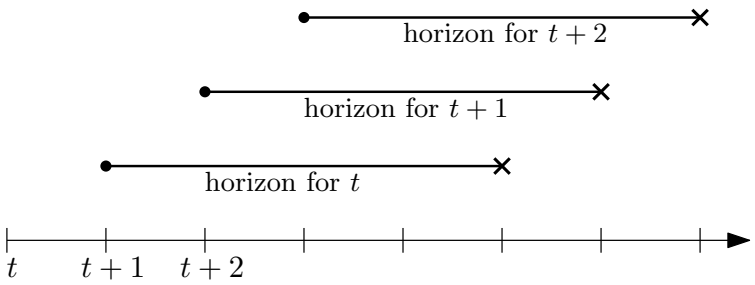


Figure 2: Example of rolling plan at each month with a horizon of four months. Each plan horizon concerns incremental goals in the light of previously settled plan goals and revealed outcomes. A more realistic plan horizon of, say, twenty months could take into account seasonal variations.

Shadow prices

Resource constraints enforce choices between things that cannot be added up. Should an investment project building a flood barrier use ten thousand tonnes of steel, or a mix of steel and cement? Should a power plant burn imported gas or draw down a limited domestic reserve? Without some common unit for comparing them, boards and enterprises are simply guessing. In 1920, the Austrian economist Ludwig von Mises anticipated that the lack of an appropriate costing metric would result in the squandering of resources under planning.[49] Some two decades later, Novozhilov came to the same conclusion working from within the Soviet economy. He also realized that direct cost-based pricing fails to account for the overall cost of using a scarce input in one place as compared to what it could have produced elsewhere.

Novozhilov's perspective was that one needs to take into account the costs of economic opportunity in the planned sector as a whole. Labor is the *ultimate* production input, since it can be mobilized to produce, say, power or steel plants, but the latter cannot be mobilized to increase the amount of available labor. From both a normative and economic standpoint, collective needs should be met without squandering the finite human labor time of society. From this principle, Novozhilov was able to derive the corresponding *shadow prices* of the resources used and produced: their marginal labor costs.[50]

To explain what this means, suppose for a given future month the committed final orders for electrical power were 100 GWh. Then the shadow price per GWh of electrical power supply is the *additional* amount of worker-hours in the entire planned sector required to increase the order by one GWh. This pricing is applicable to all internal resources, whether it is tons of steel or hours of emergency care work. On the other hand, suppose for the same month the amount of crude oil imports must be capped to 200 kilotonnes. Then the shadow price per kilotonne of imported crude oil is the amount of worker-hours that could be *saved* by increasing the imports by one unit. This gives investment and provision boards a metric for evaluating heterogeneous resources against one another in human labor time, without first having to settle on countless arbitrary conversions between limited quantities of, say, freshwater withdrawals, land use or mineral extraction that would obscure actual physical constraints.[51]

[49] See his critique of Neurath's physical in-kind planning in Mises (1920). The problem of economic *calculation* is therefore not reducible to a matter of computation.

[50] Building on Kantorovich's pioneering work, Novozhilov derived them as the Lagrange multipliers, or dual variables, of a counterfactual labor minimization problem; see Novozhilov (1970).

[51] As von Mises anticipated, this is a difficulty for any in-kind accounting scheme that aggregate heterogeneous physical measures to a few metrics for the purpose of allocative decisions: since they are non-invertible, provide no general distinction between different underlying compositions of resource use. Suppose, for illustration, that resource use was reported in a single index of 'metric tonnes' across all materials. Consider two projects each budgeted at 10,000 tonnes: a flood barrier requiring 9,000 tonnes of sand and gravel plus 1,000 tonnes of steel, and a battery plant requiring 1,000 tonnes of cobalt plus 9,000 tonnes of aluminum. The metric cannot distinguish

The idea of using shadow prices to guide a planned economy motivated several reform proposals in the Soviet-type economies.[52] They would nevertheless stumble upon the information and incentive problems discussed above: acquiring and processing the fine-grained, continually changing information that the models required as well as aligning the interests of enterprises asked to report it truthfully. We propose generalizing Novozhilov's approach in two respects, addressing both of these in turn. First, the shadow prices are derived for the incremental use of resources over a rolling plan horizon. The monthly shadow prices of, say, local water from a limited fresh water supply will vary over the plan horizon depending on how its scarcity impinges on the productive capacity of the planned sector.

Second, the shadow prices take into account irreducible uncertainties from harvests, extraction yields or delivery times. This is made possible using the public database populated by enterprises' autonomous orders and inventory adjustments. By comparing ex-ante orders and ex-post outcomes in the past, deviations in the anticipated data can be tracked for each enterprise. For instance, a power plant may have delivered output with a 5% shortfall, or an aluminium smelter may have used 3% more power than its ex-ante consumption, and so on. The shadow prices using uncertain data are derived by solving *counterfactual labor-saving problem* which seeks to find the *minimal* amount of labor under unfavorable deviations in the enterprises' anticipated data.[53] As a consequence, outputs which can be produced *reliably* will, all else equal, receive *lower* shadow prices since they require less labor to compensate for potential deviations from the ex-ante orders.

A coordination agency can publish a monthly list of shadow prices over all resources by solving a large-scale optimization problem for each new plan horizon. Since each enterprise uses only a small subset of possible inputs to produce a small set of outputs, the resulting computational problem is sparse and therefore highly scalable.[54] Budget-constrained sectoral boards and enterprises can place orders based on these opportunity costs measured in worker-hours.

The total budget for the sectoral boards can be formed as:

$$\text{Final demand budget} = \text{Labor-time available} + \text{Shadow price of resource constraints},$$

based on the employable labor available in the planned sector and the specified constraints on crude oil imports, area of tree felling, and so on. This budget is allocated to the boards in proportions stipulated by the elected economic plan. It will therefore constrain final demand for, say, electrical power and insulin, ordered through granted provision projects. For investment projects that grant the construction of new assets, such as a new custom silo, boards must not only consider the direct labor time and the worker-hours price of inputs, but also the additional labor for maintaining and repairing investment goods over its estimated operational life. This must be covered by a granted project, which the board periodically reviews and can choose to renew or terminate.

between them, yet if cobalt imports were restricted to, say, 1,200 tonnes a year, approving the battery plant alone would exhaust most the imports, foreclosing every other cobalt-dependent project. A shadow price, by contrast, would reveal the cost immediately, since cobalt's binding scarcity would set its price orders of magnitude above sand's. Thus shadow prices provide an economic rationale for decisions that are always 'multi-criterial', cf. Benanav (2025a).

[52] See Sutela (1991) and Ericson (2019). There was a striking lack of consensus about which social objective function to ground the shadow prices on.

[53] The problem can be formalized as a robust linear optimization problem using the dual formulation. An outline is provided in the appendix, see Bertsimas and Sim (2004) for details.

[54] Large-scale sparse linear optimization problems have been actively researched for decades and recent methods can solve problems involving up to one billion variables to suitable accuracy within twenty hours using commercially available hardware. Thus runtimes of "a thousand years" cited by Sunkara et al. (2025) are not grounded in research on sparse linear programs. See Luenberger and Ye (2021) and Applegate et al. (2026).

Shadow prices give boards and enterprises a common metric for allocative decisions on common resources, but nothing yet ensures that the resulting final orders are jointly feasible with overall resource constraints.

Collective goal balancing

Suppose a month in a plan horizon calls for 10 GWh of electricity and a 20 kilotonne cap on imported oil. Are the two even compatible – can the economy deliver both? If not, how should it be tackled? Soviet-type planning confronted such feasibility issues through the annual method of material balances, iteratively adjusting production targets until reported input requirements and supplies for thousands of product categories appeared to reconcile on paper. But as noted above, the enterprises supplying the information had incentives to understate their true capacity and report inflated input requirements, so that balances left too little slack to hold in practice.[55]

Before final orders with the enterprises can be settled along with resource constraints for a new plan horizon, both must be rendered jointly feasible. This can be posed as a *goal-balancing problem*, using the enterprises' supply and use data from the previously settled plan horizon. If the incremental plan goals in a new plan horizon are found jointly feasible, then final orders are placed with the enterprises as is. On the other hand, if these goals conflict, then the resource constraints and the final demand must be rebalanced. We suggest rebalancing them such that the sum of percentage deviations from the initial goals (10 GWh, 20 kilotonnes, and so on) is *minimized*.[56] The goal-balancing problem can be posed as another large-scale optimization problem, which arrives at a set of final orders and resource constraints that are jointly feasible with high probability when taking into account deviations in anticipated orders. This approach automatically builds in adaptable slack and minimizes problems of unrealizable plan goals that plagued the Soviet-type economies.

Suppose this goal balancing solution adjusts the initial goals to 9 GWh of power and 21 kilotonnes of oil imports for feasibility. Then the projects can place their final orders accordingly. On the other hand, when urgent it can be flagged for renegotiation between the sectoral boards and the viability assembly, where the ultimate decision power rests with the latter. The goal-balancing method thereby forms a bridge between the limited perceptive capacities of decision-makers at the sectoral boards and the vast complexities of joint production across the planned sector. It avoids choking the planning process to decide over details[57], yet enables the exploration of production possibilities; the elucidation of interdependencies between conflicting goals; and the improvement in efficiency.[58] The goal-balancing method realizes Neurath's vision of planning in physical quantities at a stage in planning when goals must be made jointly feasible.[59]

Enterprise budgets

An enterprise under Soviet-type planning facing an unfavorable allocation from the center had every incentive to report inflated requirements. Even when falling short of its production targets, a wasteful enterprise could routinely renegotiate its allocation after the fact, because the center was reluctant to wind it down. This resulted in what Kornai identified as an ineffective budget constraint.[60] The resulting squander of resources prevented collective goals from being satisfied

[55] See Nove (1986) and Kornai (1980).

[56] Thus autonomous regulation is by itself ex-post adjustment that corrects short-run disturbances, whereas the physical bounds are determined in anticipation of use by goal balancing. This contrast to market ex-post adjustment discussed above.

[57] That is the meetings problem that bedevils proposals such as Albert and Hahnel (1991).

[58] Paraphrasing Kornai (1970), who was deeply involved in trying to improve the planning system in Hungary.

[59] It also avoids the economically rather unmotivated objective function initially proposed by Kantorovich – repeated also in Cockshott et al. (2022) and Dapprich (2023) – in which final outputs are evaluated in fixed, specified proportions, see Barden (1975).

[60] Kornai (1980).

even where they were, in principle, feasible. We propose instead that enterprises face effective budget constraints derived directly from the goal-balancing solution, rather than negotiated after the fact.

The goal-balancing solution produces, as an important by-product, a robust estimate of the incremental labor and non-labor requirements of each enterprise across the new plan horizon. We propose a corresponding *allocation of operational budgets* to enterprises by the coordination agency. Suppose that for a given month, a steel mill's estimated labor requirement is 100 worker-hours, and the shadow-price value of its non-labor requirements comes to 200 worker-hours. For the new plan horizon it may then autonomously employ additional labor and place internal orders within an allocated labor budget of 100 worker-hours and non-labor budget of 200 worker-hours, respectively. All else equal, the goal-balancing solution allocates larger budgets to enterprises whose outputs are in demand and that contribute to making the plan goals maximally feasible.

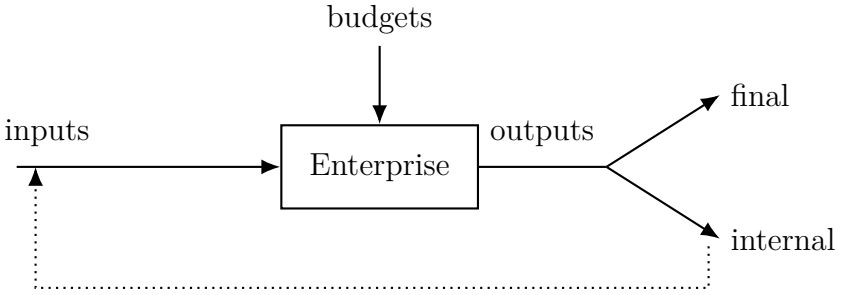


Figure 3: Each enterprise is granted budgets that bound the inputs and labor it may employ. Its outputs are placed as final orders under granted projects or as internal orders to other enterprises, which recursively become inputs elsewhere in the planned sector.

Note that the budgets serve only the purpose of resource allocation, not exchange: when an enterprise places an order, its non-labor budget is debited accordingly, but the enterprise fulfilling that order does not thereby receive a transferable credit of its own – its budgets were already set independently by the same goal-balancing process, based on its own anticipated requirements. Credits are therefore cleared, not transferred: an enterprise's operational budget is simply consumed as its ceiling is reached. Unused operational budgets cannot be hoarded. Expensive investment goods – for instance a new custom silo for a cement producer – are instead funded by projects granted by a sectoral investment board.

With an allocated labor budget, an enterprise may increase its employment of labor, but it does not pay for that labor's remuneration out of any revenue of its own. Instead, its workers receive monetary compensation for their effort – which they may spend freely in the market sector – from a sovereign *labor fund* managed by *sectoral labor boards*. By this separation, the enterprise's budget governs only how much labor *time* it may employ, not what that time is *compensated*.

Compensation for work is a necessary, if insufficient, component of worker motivation on a mass scale. Its role is to mobilize labor towards realizing investment and provision projects across the planned sector. Labor *shortages* inevitably constrain the planned sector's capacity, since the transfer and training of workers can never exactly match the shifting demand for particular skills; even where shortages in, say, mines or hospitals could eventually be overcome by training, reorganization or automation, this would not resolve the problem in the *short run*. Differentials in compensation – in both remuneration and vacation time – are therefore an indispensable instrument for addressing labor shortages over shorter horizons. Since workers can freely apply for positions across enterprises, compensation differentials become a central instrument for addressing sectoral shortages while avoiding the restricted labor mobility characteristic of Soviet-type economies. The sectoral labor boards credit work performed in enterprises at these varying rates from the labor fund, whose overall size is fundamentally constrained by the need to

maintain consumer price stability and a balance of payments within limits set by the viability assembly.[61] The politically acceptable levels of compensation differentials and shortages are a trade-off to be periodically renegotiated through these sectoral boards, which include worker representatives to defend their interests.[62] In sum, workers have an interest in maximizing the labor budget allocated to their own enterprise, since this secures their employment. But the level of compensation is set independently, through the labor fund. Whether this separation is enough to align enterprises' and workers' interests with collective goals more broadly is the question we turn to next.

Incentive structures

As noted above, in Soviet-type planning the center depended on the very enterprises it sought to regulate to supply necessary information in the first place. Under ineffective budget constraints, enterprises had little reason to report it correctly, since inflated requirements and understated capacity would be rewarded rather than corrected. Setting in place an effective budget constraint removes the enterprise's ability to renegotiate allocations after the fact, but it does not, by itself, remove its interest in masking the squander or hoarding of resources within that harder constraint. Crucially, such incentives *prevent collective goals from being satisfied*. We now show how shadow prices, goal balancing and enterprise budgets, taken together, close this remaining gap.[63]

Using the three instruments, anticipatory regulation operates as a recurring monthly procedure, overseen by a public coordination agency:

1. *Shadow prices* of resources are announced for the new plan horizon of, say, twenty months.
2. The budget for final orders over this horizon is *divided between the sectoral boards* for investment and provision in accordance with the elected economic program.
3. The final orders and resource constraint goals are *balanced*, taking into account internal resource usages by enterprises.
4. Enterprises receive *budget allocations* with which they can employ labor and place new internal orders they deem necessary.
5. As the orders are placed, the plan horizon is *settled*, and a subsequent one begins in the next month.

In the interest of remaining operational, enterprises prefer larger budget allocations. Suppose, for instance, that a steel producer hoards production inputs, or inflates its labor inputs, without a corresponding increase in steel deliveries. The increased resource usage to produce the same output raises the shadow prices of its steel products, which diminishes incoming orders – projects and other enterprises will prefer better-costed steel products elsewhere. This *reduces* its subsequent budget allocation. Moreover, if the increased usage presses against resource constraints, the goal-balancing solution will reallocate budgets to more efficient steel producers. On the other hand, running down inventories and buffer capacities to appear more efficient would be counterproductive, since any resulting shortfall in deliveries would have the *same* effect: additional labor would be needed to compensate for such deviations and budgets must be reallocated to more producers with more reliable outputs. We conclude that shadow prices, and

[61] The size of the labor fund is therefore constrained by the level of taxation and sales of essential products to the market sector.

[62] This would make negotiations not too unlike the peaks of sectoral bargaining in countries like Sweden and Norway. The viability assembly could possibly define limits on the differentials as well, but their political salience will depend on the basic needs secured through public provision.

[63] Analysis of incentive compatibility is rather uncommon among socialist proponents of planning; see Laibman (2011) for an exception.

budget allocations based on a goal-balancing solution, incentivize enterprises to utilize collective resources *efficiently* in order to meet their orders *reliably*.

The *degree* of interest alignment depends to what extent an enterprise's outputs are substitutable, and whether they can be produced by more than one enterprise. Table 1 illustrates the four resulting cases with concrete examples.

	Multiple enterprises	Single enterprise
High substitutability	Steel producers	Fertilizer plant
Low substitutability	Power plants	Insulin producer

Table 1: Interest alignment across four cases, classified by output substitutability and whether it is produced by multiple enterprises or just one.

For steel producers or a single fertilizer plant, whose outputs can be substituted by alternatives, a change in shadow price has a significant effect on demand. This reinforces a high degree of interest alignment: demand will switch towards less costly alternatives. Regional power plants or a sole insulin producer, by contrast, supply products with few or no substitutes, so changes in their shadow prices have much weaker effects on demand in the short run. This would be a rather common situation, since the planned sector is focused on essential and vital products. Among regional power plants, however, the goal-balancing solution still allocates budgets according to how each plant contributes to making final demand and resource constraints jointly feasible. A more resource-efficient plant would receive a larger operational budget, which ensures a degree of interest alignment.

The main case for concern is therefore the sole insulin producer: an enterprise supplying a nonsubstitutable good with no alternative producer at all. If left unchecked, such an enterprise could squander labor and hoard resources with comparatively little consequence. It is therefore advisable to assign sectoral investment boards the task of investing in parallel production capacities for goods like insulin or cement, converting a case of unchecked monopoly into one where interests are once again aligned through comparison across enterprises.

In summary, the budget allocation mechanism generates an imperative for enterprises to find and meet demand for their outputs in a resource-efficient and reliable manner in order to secure their budget allocation. This incentivizes *dynamic efficiency*, both through the refinement of existing production methods and the innovation of new, useful outputs. The mechanism is compatible with competitive as well as cooperative interactions between enterprises. Unlike in the Soviet-type economies, enterprises that persistently fail to meet this imperative are wound down within the planned sector – either through steadily diminishing budget allocations, reflecting a lack of demand or efficiency, or through the non-renewal of investment projects. While workers at a wound-down enterprise risk losing that particular employment, they do not risk losing their livelihood, since the labor fund can be used to cover for reskilling and the broader provision secured through public and market-sector goods.

Conclusions

Every generation of socialists has had to answer the same question in different terms: deliver what, exactly, and how? A feasible socialism in the 1930s had to deliver full employment and modernizing investment amid capitalist stagnation; in the 1960s, gains beyond those already won by organized labor under state-led growth. Ours must answer to the tightening constraints of climate change, geopolitical conflict and demographic decline – what we have called the Viable

Allocation Problem: the capacity to efficiently secure the resources required for a population's well-being, while limiting the extraction of critical resources and the accumulation of waste.

We have argued that market allocation, whether carried out by capitalist or collectively-owned firms, cannot solve this problem: it requires firms to disclose the very information their income depends on withholding, leaving any regulatory body to act on partial and distorted information. Even if investments were socialized, the corrective mechanism of competitive markets operates only after a bad investment has already been made, so resources are squandered and stipulated limits exceeded before it is weeded out. Addressing the Viable Allocation Problem therefore motivates restoring planning to the *center* of socialist economics.

Building on ideas from economists deeply engaged with the practical problems of Soviet-type planning, we have proposed a minimal institutional structure in which a planned sector produces the bulk of essential and vital products – and subordinates a market sector concerned with private consumption. Following Lange, a democratic plan for economic development is set through sectoral allocations for investments and provision, without the need for a center to engage in production details. Following Kornai's insights, enterprises are left to regulate their own production autonomously, adjusting to orders, inventories and local information. This can sustain production while absorbing shocks without central instructions or market prices. Following Novozhilov's insights, allocative decisions are guided by shadow prices, a costing metric denominated in worker-hours, which give enterprises a rational basis for valuing scarce resources without requiring market exchange. Combined within a rolling planning procedure of shadow pricing, budget allocation, and the balancing of final demand against resource constraints, these three insights ground a complementary system of *anticipatory* regulation.

This addresses the Viable Allocation Problem directly. Goal balancing keeps final orders and enterprise production consistent with the boundaries set collectively, while shadow prices and budget allocations give enterprises a genuine incentive to meet demand efficiently and reliably – without requiring market exchange, the private appropriation of enterprise revenue, or a central authority able to observe what only enterprises themselves can know.

With this contribution, we hope to spark a debate among socialist economists.[64] Not another one about the virtues and vices of markets versus plans, but rather *how* planning should operate in a socialist economy that enables the *flourishing* of individual and collective life *within* our planetary boundaries.

References

Seth Ackerman. Socialism Can Work. Cybersocialism Can't. Forthcoming, 2026.

Michael Albert and Robin Hahnel. *The Political Economy of Participatory Economics*. Princeton University Press, Princeton, NJ, 1991. doi: 10.1515/9780691216003.

Robert C. Allen. *Farm to Factory: A Reinterpretation of the Soviet Industrial Revolution*. Princeton University Press, Princeton, NJ, 2003.

David Applegate, Mateo Díaz, Oliver Hinder, Haihao Lu, Miles Lubin, Brendan O'Donoghue, and Warren Schudy. PDLP: A practical first-order method for large-scale linear programming. *Mathematical Programming Computation*, 2026. doi: 10.1007/s12532-026-00309-2.

Drew S. Barden. Optimal planning: A radical critique. *Review of Radical Political Economics*, 7(4):33–47, 1975. doi: 10.1177/048661347500700403.

Mike Beggs. The market and workplace in a democratic socialism. *Catalyst*, 6(3), 2022. URL https://catalyst-journal.com/2023/01/the-market-and-workplace-in-a-democratic-socialism.

[64] Research into these questions is also being pursued within the recently formed International Network for Democratic Economic Planning (INDEP), see https://www.indep.network/.

Mike Beggs, Ben Burgis, and Bhaskar Sunkara. *The Blueprint: How Socialism Can Work in the Real World*. Verso, London, 2026.

Aaron Benanav. Beyond capitalism—1. *New Left Review*, (153), 2025a. URL https://newleftreview.org/issues/ii153/articles/aaron-benanav-beyond-capitalism-1.

Aaron Benanav. Beyond capitalism—2. *New Left Review*, (154), 2025b. URL https://newleftreview.org/issues/ii154/articles/aaron-benanav-beyond-capitalism-2.

Dimitris Bertsimas and Melvyn Sim. The price of robustness. *Operations Research*, 52(1):35–53, 2004. doi: 10.1287/opre.1030.0065.

John Burnheim. *Is Democracy Possible? The Alternative to Electoral Politics*. Polity Press, Cambridge, 1985.

W. Paul Cockshott and Allin F. Cottrell. *Towards a New Socialism*. Spokesman, Nottingham, UK, 1993. URL https://users.wfu.edu/cottrell/socialism_book/.

W. Paul Cockshott and David Zachariah. Hunting productive work. *Science & Society*, 70(4):509–527, 2006.

W. Paul Cockshott, Allin Cottrell, and J. Philipp Dapprich. *Economic Planning in an Age of Climate Crisis*. Independently published, 2022. ISBN 9798360125563. URL https://westminsterresearch.westminster.ac.uk/item/x07v6/economic-planning-in-an-age-of-climate-crisis.

G. A. Cohen. *Why Not Socialism?* Princeton University Press, Princeton, NJ, 2009.

Jan Philipp Dapprich. Optimal planning with consumer feedback: A simulation of a socialist economy. *Review of Political Economy*, 35(4):1136–1156, 2023. doi: 10.1080/09538259.2021.2005367.

Pat Devine. *Democracy and Economic Planning: The Political Economy of a Self-Governing Society*. Polity Press, Cambridge, UK, 1988.

Maurice Dobb. *Socialist Planning: Some Problems*. Number 12 in Socialism Today. Lawrence & Wishart, London, 1970.

David A. Dyker. *The Future of the Soviet Economic Planning System*. Croom Helm, London, 1985.

John Eatwell, Murray Milgate, and Peter Newman, editors. *Problems of the Planned Economy*. The New Palgrave. Palgrave Macmillan, London, 1990.

Michael Ellman. *Planning Problems in the USSR: The Contribution of Mathematical Economics to their Solution 1960–1971*. Cambridge University Press, Cambridge, 1973.

Michael Ellman. The fundamental problem of socialist planning. *Oxford Economic Papers*, 30(2):249–262, 1978.

Michael Ellman. *Socialist Planning*. Cambridge University Press, Cambridge, 3rd edition, 2014.

Richard E. Ericson. The classical Soviet-Type economy: Nature of the system and implications for reform. *Journal of Economic Perspectives*, 5(4):11–27, 1991. doi: 10.1257/jep.5.4.11.

Richard E. Ericson. The growth and marcescence of the "system for optimal functioning of the economy" (sofe). *History of Political Economy*, 51, 2019. doi: 10.1215/00182702-7903276. Annual supplement.

Sam Gindin. Socialism for realists. *Catalyst: A Journal of Theory and Strategy*, 2(3), 2018. URL https://catalyst-journal.com/2018/12/socialism-for-realists.

Paul R. Gregory. The Stalinist command economy. *The ANNALS of the American Academy of Political and Social Science*, 507(1):18–25, 1990. doi: 10.1177/0002716290507001002.

Paul R. Gregory and Joel Sailors. The Soviet Union during the Great Depression: The autarky model. In Theo Balderston, editor, *The World Economy and National Economies in the Interwar Slump*, pages 191–210. Palgrave Macmillan, Basingstoke, 2003.

Gregory Grossman. Notes for a theory of the command economy. *Soviet Studies*, 15(2):101–123, 1963. doi: 10.1080/09668136308410352.

Max Grünberg. Automating away the centre? optimal planning and the menace of bureaucratisation. *Competition & Change*, 29(1):38–63, 2025. doi: 10.1177/10245294231218868.

Simon Hannah. *Reclaiming the Future: A Beginner's Guide to Planning the Economy*. Pluto Press, London, 2024. ISBN 9780745350202.

Mark Harrison. Industrial mobilisation for World War II: A German comparison. In John Barber and Mark Harrison, editors, *The Soviet Defence Industry Complex from Stalin to Khrushchev*, pages 99–117. Macmillan Press, Basingstoke, 2000.

Mark Harrison. The fundamental problem of command: Plan and compliance in a partially centralised economy. *Comparative Economic Studies*, 47(2):296–314, 2005. doi: 10.1057/palgrave.ces.8100110.

Friedrich A. Hayek. The use of knowledge in society. *American Economic Review*, 35(4):519–530, 1945.

Jason Hickel and Dylan Sullivan. How much growth is required to achieve good lives for all? insights from needs-based analysis. *World Development Perspectives*, 35:100612, 2024. doi: 10.1016/j.wdp.2024.100612.

Leonid Hurwicz. The design of mechanisms for resource allocation. *American Economic Review*, 63(2): 1–30, 1973.

Alberto Isidori. The internal model principle – the seminal contributions of Francis and Wonham. *IEEE Control Systems Magazine*, 2025. Exact volume/issue/page numbers and DOI not independently verified.

Leif Johansen. *Lectures on Macroeconomic Planning, Part 1: General Aspects*. North-Holland, Amsterdam, 1977.

Leif Johansen. *Lectures on Macroeconomic Planning, Part 2: Centralization, Decentralization, Planning under Uncertainty*. North-Holland, Amsterdam, 1978.

Gergely Kőhegyi and János Mátyás Kovács. Mathematical economics outside the neoclassical paradigm? evolution of planning concepts in Hungary under communism. In János Mátyás Kovács, editor, *Communist Planning versus Rationality: Mathematical Economics and the Central Plan in Eastern Europe and China*, pages 145–210. Lexington Books, 2022.

János Kornai. A general descriptive model of planning processes. *Economics of Planning*, 10(1):1–18, 1970. doi: 10.1007/BF00347043.

János Kornai. *Economics of Shortage*. North-Holland, Amsterdam, 1980.

János Kornai and Béla Martos. Autonomous control of the economic system. *Econometrica*, 41(3): 509–528, 1973.

János Kornai and Béla Martos, editors. *Non-Price Control*. Number 133 in Contributions to Economic Analysis. North-Holland, Amsterdam, 1981.

Maximilian Kotz, Anders Levermann, and Leonie Wenz. The economic commitment of climate change. *Nature*, 628(8008):551–557, 2024. doi: 10.1038/s41586-024-07219-0.

David Laibman. Incentive design, iterative planning and local knowledge in a maturing socialist economy. *International Critical Thought*, 1(1):1–22, 2011. doi: 10.1080/21598282.2011.566039.

Oskar Lange. The role of planning in socialist economy. *Indian Economic Review*, 4(2):1–15, 1958.

Frédéric Legault and Simon Tremblay-Pepin. A brief sketch of three models of democratic economic planning. Technical Report 2, Élisabeth-Bruyère School of Social Innovation, Saint Paul University, 2021. URL https://innovationsocialeusp.ca/wp-content/uploads/2021/04/Note-2-Legault-and-Tremblay-Pepin-Democratic-Planning.pdf.

David G. Luenberger and Yinyu Ye. *Linear and Nonlinear Programming*, volume 228 of *International Series in Operations Research & Management Science*. Springer, Cham, 5th edition, 2021. doi: 10.1007/978-3-030-85450-8.

Erik Lundberg. Ingvar Svennilson: A note on his scientific achievements and a bibliography of his contributions to economics. *The Swedish Journal of Economics*, 74(3):313–328, 1972.

Andreas Malm. *How to Blow Up a Pipeline: Learning to Fight in a World on Fire*. Verso, London, 2021.

Michael Manove. A model of Soviet-Type economic planning. *American Economic Review*, 61(3):390–401, 1971.

Béla Martos. Control and coordination of economic activity. In John Eatwell, Murray Milgate, and Peter Newman, editors, *Problems of the Planned Economy*, The New Palgrave. Palgrave Macmillan, London, 1990. doi: 10.1007/978-1-349-20863-0_9.

Nikolai Matni, Aaron D. Ames, and John C. Doyle. A quantitative framework for layered multirate control: Toward a theory of control architecture. *IEEE Control Systems Magazine*, 44(3):52–94, 2024. doi: 10.1109/MCS.2024.3382388.

Ludwig von Mises. Die wirtschaftsrechnung im sozialistischen gemeinwesen. *Archiv für Sozialwissenschaften und Sozialpolitik*, 47:86–121, 1920. Translated as "Economic Calculation in the Socialist Commonwealth," in F.A. Hayek (ed.), *Collective Economic Planning*, Routledge, 1935.

John Michael Montias. Planning with material balances in Soviet-Type economies. *American Economic Review*, 49(5):963–985, 1959.

Otto Neurath. *Durch die Kriegswirtschaft zur Naturalwirtschaft*. Georg D. W. Callwey, Munich, 1919.

Alec Nove. *The Soviet Economic System*. Unwin Hyman, London, 3rd edition, 1986.

Alec Nove. *The Economics of Feasible Socialism Revisited*. Routledge, London, 2nd edition, 1991.

Alec Nove. *An Economic History of the USSR, 1917–1991*. Penguin Books, London, 3rd edition, 1992.

Viktor V. Novozhilov. *Problems of Cost-Benefit Analysis in Optimal Planning*. International Arts and Sciences Press, White Plains, NY, 1970. Translation of the 1967 Russian original.

Leigh Phillips and Michal Rozworski. *The People's Republic of Walmart: How the World's Biggest Corporations are Laying the Foundation for Socialism*. Verso, London, 2019.

Adam Przeworski. *Capitalism and Social Democracy*. Studies in Marxism and Social Theory. Cambridge University Press, Cambridge, 1985.

Roy Radner. Decentralization and incentives. In Theodore Groves, Roy Radner, and Stanley Reiter, editors, *Information, Incentives, and Economic Mechanisms*, pages 3–47. University of Minnesota Press, Minneapolis, MN, 1987.

John E. Roemer. *A Future for Socialism*. Harvard University Press, Cambridge, MA, 1994.

John E. Roemer. Socialism revised. *Philosophy & Public Affairs*, 45(3):261–315, 2017. doi: 10.1111/papa.12089.

David Schweickart. *After Capitalism*. Rowman & Littlefield, Lanham, MD, 2002.

David Schweickart. Nonsense on stilts: Michael Albert's Parecon, 2006. URL http://dschwei.sites.luc.edu/parecon.pdf.

Herbert A. Simon. On the application of servomechanism theory in the study of production control. *Econometrica*, 20(2):247–268, 1952.

Leon Smolinski. Lenin and economic planning. *Studies in Comparative Communism*, 2:96–114, 1969. doi: 10.1016/S0039-3592(69)80005-0.

Piero Sraffa. *Production of Commodities by Means of Commodities: Prelude to a Critique of Economic Theory*. Cambridge University Press, Cambridge, 1960.

S. Steblev and G. Kallis. Planning and autonomous control: Reappraising the cases for economic planning in socialism. REAL – A Post Growth Deal Project Working Paper 1, Institut de Ciència i Tecnologia Ambientals (ICTA-UAB), 2025.

Bhaskar Sunkara, Mike Beggs, and Ben Burgis. The plans that failed. *Catalyst*, 9(2-3), 2025. URL https://catalyst-journal.com/2025/12/the-plans-that-failed.

Pekka Sutela. *Economic Thought and Economic Reform in the Soviet Union*. Cambridge University Press, Cambridge, 1991.

Ingvar Svennilson. *Ekonomisk planering: Teoretiska studier*. Almqvist & Wiksell, Uppsala, 1938. Doctoral thesis, Stockholm University.

Ingvar Svennilson. Planning in a market economy. *Weltwirtschaftliches Archiv*, 95, 1965. URL https://www.jstor.org/stable/40436352.

Göran Therborn. *European Modernity and Beyond: The Trajectory of European Societies, 1945–2000*. Sage, London, 1995.

Thomas E. Uebel. Incommensurability, ecology, and planning: Neurath in the socialist calculation debate, 1919–1928. *History of Political Economy*, 37(2):309–342, 2005. doi: 10.1215/00182702-37-2-309.

Adaner Usmani. Democracy and the class struggle. *American Journal of Sociology*, 124(3):664–704, 2018. doi: 10.1086/700235.

David Van Reybrouck. *Against Elections: The Case for Democracy*. The Bodley Head, London, 2016. Translated by Liz Waters.

Troy Vettese and Drew Pendergrass. *Half-Earth Socialism: A Plan to Save the Future from Extinction, Climate Change and Pandemics*. Verso, London, 2022.

Nicholas Vrousalis. Socialism unrevised: A reply to Roemer on Marx, exploitation, solidarity, worker control. *Philosophy & Public Affairs*, 49(1):78–109, 2021. doi: 10.1111/papa.12183. Cited in draft as "(2020)"; journal issue confirmed as Winter 2021.

Isabella M. Weber, Jayati Ghosh, Sophie van Huellen, Merle Schulken, and Sudeep Jain. Food price stabilization in an age of overlapping emergencies: The case for multilevel buffer stocks. Working Paper 634, Political Economy Research Institute, University of Massachusetts Amherst, 2026.

Martin L. Weitzman. Prices vs. quantities. *The Review of Economic Studies*, 41(4):477–491, 1974. doi: 10.2307/2296698.

Norbert Wiener. *Cybernetics: Or Control and Communication in the Animal and the Machine*. MIT Press / John Wiley & Sons, Cambridge, MA / New York, 1948.

David Zachariah and Paul Cockshott. Classical labor values: Properties of economic reproduction. *World Review of Political Economy*, 11(3):388–414, 2020.

Appendix

The counterfactual labor-saving problem

Consider a recent plan horizon of T months and N enterprises in which orders and employment has been contracted. Let $a_i(t) \geq 0$ denote (counterfactual) activity level of enterprise i in month t in the horizon. Activity level $a_i(t) = 1$ means that the anticipated production is left unchanged. For notation convenience, we stack all monthly activity levels into a single $NT \times 1$ vector:

$$x = \Big[a_1(1) \cdots a_N(1) \cdots a_1(T) \cdots a_N(T)\Big]^\top.$$

From the public database, we have the following set of anticipated resource usages: ℓ as a vector of anticipated labor inputs, so that $\ell^\top x$ is the total labor employed across the planned sector over the given horizon. Let R and F be the (sparse) matrices of anticipated resource use and net output, respectively, so that Rx and Fx is the total resource usages and net outputs over the horizon. Specifically, $F = B - A$, where B and A are the matrices of anticipated outputs and inputs across all enterprises over each month in the plan horizon. The settled plan goals formalized as corresponding vectors of resource constraints r and final outputs f, respectively.

We can now pose the counterfactual labor-saving problem as the following linear minimization problem:[65]

$$\begin{aligned} \min_{x \geq 0,\, \tau} \quad & \tau \\ \text{s.t.} \quad & \ell^\top x \leq \tau && (1) \\ & Rx \leq r && (2) \\ & Fx \geq f && (3) \end{aligned}$$

It rebalances all activity levels across enterprises and time to arrive at the minimal expenditure of labor to meet the goals r and f. By considering the dual of this problem, we can now turn to deriving the corresponding shadow prices.

Shadow prices: Define a collection of dual variables: $\mu \geq 0$ to (1), $\rho \geq 0$ to (2), and $\lambda \geq 0$ to (3). Stationarity of the Lagrangian in τ forces $\mu = 1$: the labor constraint's own shadow price is pinned to unity, which is precisely why ρ and λ come out *denominated in worker-hours* rather than in some arbitrary unit. Stationarity in x, together with complementary slackness on $x \geq 0$, then gives

$$\ell + R^\top \rho - F^\top \lambda \;\geq\; 0,$$

with equality on every enterprise-month actually operated ($x_i > 0$).[66] The dual problem is

$$\max_{\rho, \lambda \geq 0} \quad \lambda^\top f - \rho^\top r \quad \text{s.t.} \quad \ell + R^\top \rho \geq F^\top \lambda,$$

so ρ is the marginal labor cost of tightening a resource constraint by one unit, and λ the marginal labor saved by relaxing a final-order commitment by one unit – exactly the shadow prices described in the main text. This is a generalization of Novozhilov's formulation that explicitly includes intermediate products via F.

Shadow prices that take into account uncertainty. The anticipated data in ℓ, R and F is subject to deviations from their true, ex-post realizations described in the main text (a power plant's

[65] The auxiliary variable τ is not strictly needed to minimize labor – at the optimum it simply equals $\ell^\top x$ – but writing the objective in this epigraph form is what lets the robust version below bound labor *uniformly* against every possible realization of the true, uncertain coefficients.

[66] A enterprise-month is thus only active if its labor cost plus the shadow value of the resources it draws down, $\ell + R^\top \rho$, is covered by the shadow value of what it delivers, $F^\top \lambda$ – a break-even condition in worker-hours.

shortfall, a smelter's excess draw, etc.). Solving (1)–(3) against these point estimates alone would understate the labor actually needed whenever realizations turn out worse than anticipated, and the resulting plan could fail to be feasible once those deviations materialize.

Following Bertsimas and Sim (2004), we introduce ambiguity sets as a budget-constrained polytope around the nominal data:

$$\ell^\top x \le \tau \;\; \forall\, \ell \in \mathcal{U}_\ell, \qquad Rx \le r \;\; \forall\, R \in \mathcal{U}_R, \qquad Fx \ge f \;\; \forall\, F \in \mathcal{U}_F.$$

A single row. Consider the mth row of (2), say $\sum_j R_{mj} x_j \le r_m$, and let J_m index the enterprise-months with uncertain resource coefficients in row m, so that $R_{mj} \in \mathcal{U}_{R_m} := \{R_{mj} \in [\hat{R}_{mj} - \widetilde{R}_{mj}, \hat{R}_{mj} + \widetilde{R}_{mj}]$ for $j \in J_m$, with at most Γ_m of these deviating from $\hat{R}_{mj}$ at once$\}$. Requiring $\sum_j R_{mj} x_j \le r_m$ now means protecting against the worst-case deviations in the resource usage: at most $\lfloor \Gamma_m \rfloor$ coefficients deviate to their worst-case value.[67] Bertsimas and Sim (2004) showed that the corresponding robust inequality can be parameterized linearly using auxiliary variables as:

$$\begin{aligned} \sum_j \hat{R}_{mj} x_j + \Gamma_m p_m + \sum_{j \in J_m} q_{mj} &\le r_m, \\ p_m + q_{mj} &\ge \widetilde{R}_{mj} x_j, \quad \forall j \in J_m, \\ p_m \ge 0, \quad q_{mj} &\ge 0, \quad \forall j \in J_m. \end{aligned} \tag{4}$$

Note that no exponential enumeration of worst cases is needed: row m only grows by one variable p_m and $|J_m|$ variables q_{mj}.

All three constraint families. The same construction applies row by row to (1) and (3), using corresponding ambiguity sets $\mathcal{U}_\ell$ and $\mathcal{U}_{F_k}$ with corresponding error budgets Γ^ℓ and Γ_k. For the labor inputs, with J^ℓ indexing the enterprise-months with uncertain labor coefficients and auxiliary variables $p^\ell, q^\ell_j \ge 0$:

$$\hat{\ell}^\top x + \Gamma^\ell p^\ell + \sum_{j \in J^\ell} q^\ell_j \le \tau, \qquad p^\ell + q^\ell_j \ge \widetilde{\ell}_j x_j \;\; \forall j \in J^\ell.$$

For a row k of the final-orders constraint $\hat{F}x \ge f$, the adversarial direction is a *shortfall* in delivery, so the protection term is subtracted from anticipated output rather than added to anticipated use; with J_k indexing enterprise-months with uncertain output coefficients and $p_k, q_{kj} \ge 0$:

$$\sum_j \hat{F}_{kj} x_j - \Gamma_k p_k - \sum_{j \in J_k} q_{kj} \ge f_k, \qquad p_k + q_{kj} \ge \widetilde{F}_{kj} x_j \;\; \forall j \in J_k.$$

Replacing (1)–(3) with these row-wise robust counterparts yields a single linear program of comparable size – the extra variables p, q are added once per row, not once per scenario – whose solution x is guaranteed feasible against any simultaneous deviation of up to $\lfloor \Gamma \rfloor$ coefficients per row. Its dual again yields shadow prices ρ and λ, now attached to the robust constraints (4) and its analogues, and it is this robust dual that generates the shadow prices under uncertainty described in the main text: outputs that can be produced reliably (small $\widetilde{F}_{kj}$, or produced by enterprises with slack elsewhere) draw a smaller protection term and so receive a lower shadow price than those whose delivery is uncertain.

The goal-balancing problem

Consider now a new planning horizon, using the corresponding anticipated ℓ, R and F read from the public database. The viability assembly and projects overseen by the sectoral boards

[67] And one further coefficient deviates by the fractional remainder $\Gamma_m - \lfloor \Gamma_m \rfloor$.

initially propose a new set of plan goals r° and f° – vectors of resource caps and final orders, one entry per constrained resource-month and ordered-good-month, respectively – against an initial labor limit τ°. These plan goals need not be *jointly* feasible: there may be no activity vector $x \geq 0$ that keeps resource use within r°, employs no more labor than τ°, and still meets every order in f°.[68]

To resolve this without arbitrarily favoring any one board's goals, we let the labor limit, each resource cap and each final order all be adjusted by an amount to be determined jointly. Let $w \in [0, w_{\max}]$ denote the proportional slack added to the labor limit, $u_m \geq 0$ the proportional slack added to the mth resource constraint, and $v_k \in [0,1]$ the proportional shortfall accepted on the kth final order. Suppose enterprises report their excess production capacity, so for an enterprise-month it is η_i. The goal-balancing problem is then

$$\begin{aligned}
\min_{x,\,w,\,u,\,v} \quad & w + \sum_m u_m + \sum_k v_k \\
\text{s.t.} \quad & \ell^\top x \leq (1+w)\,\tau^\circ && (5)\\
& \sum_j R_{mj} x_j \leq (1+u_m)\, r^\circ_m, \quad \forall m, && (6)\\
& \sum_j F_{kj} x_j \geq (1-v_k)\, f^\circ_k, \quad \forall k, && (7)\\
& 0 \leq w \leq w_{\max}, \qquad u_m \geq 0 \;\; \forall m, \qquad 0 \leq v_k \leq 1 \;\; \forall k \qquad 0 \leq x_j \leq (1+\eta_j) \;\; \forall j.
\end{aligned}$$

Since τ°, r°_m and f°_k are fixed constants, the right-hand sides of (5)–(7) are linear in w, u_m and v_k, so the problem remains a linear program. Minimizing the sum of relative deviations $w + \sum_m u_m + \sum_k v_k$ finds the *smallest* total rebalancing of the initial goals – including the labor limit itself – that renders them jointly achievable, rather than adjusting any single board's target by more than the plan as a whole requires.

At the optimum (x^*, w^*, u^*, v^*), define the *balanced goals*

$$\tau := (1+w^*)\,\tau^\circ, \qquad r_m := (1+u^*_m)\, r^\circ_m, \qquad f_k := (1-v^*_k)\, f^\circ_k.$$

These are, by construction, jointly *feasible*: x^* satisfies $\ell^\top x^* \leq \tau$, $\sum_j R_{mj} x^*_j \leq r_m$ for every m, and $\sum_j F_{kj} x^*_j \geq f_k$ for every k. Thus r and f now constitute the balanced goals for the new plan horizon.

Enterprise budgets. The anticipated input requirements for a particular enterprise-month can be expressed compactly using a standard basis vector e_i:

$$\ell^\top e_i x^*_i \qquad A e_i x^*_i \qquad R e_i x^*_i.$$

Thus a labor budget allocated to enterprise-month i would be $\ell^\top e_i x^*_i$. The non-labor inputs can be budgeted similarly into different categories if needed, that is, a shadow-price budget for all imported fossil fuels, fresh water usage, and so on, respectively. In this simplest case, they can be aggregated into a single category so that:

$$\textsc{Labor budget} = \ell^\top e_i x^*_i \quad \text{and} \quad \textsc{Non-labor budget} = \lambda^\top A e_i x^*_i + \rho^\top R e_i x^*_i,$$

respectively.

Robustness under uncertainty. As in the labor-saving problem, ℓ, R and F here are again anticipated rather than certain, so the same technique of Bertsimas and Sim (2004) applies directly: replacing (5)–(7) with their robust counterparts – exactly as in (4) and its analogues for ℓ and F, but with τ, r_m and f_k there replaced by $(1+w)\tau^\circ$, $(1+u_m)r^\circ_m$ and $(1-v_k)f^\circ_k$ – yields a robust goal-balancing problem, jointly optimized over x, w, u, v and the auxiliary variables p, q, that remains a single linear program.

[68] τ° may be a total labor budget set for the labor fund.